Adaptive Multipath Transport Scheduling in Heterogeneous Networks

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Proposal and Objectives

Background. TCP currently works by making connection using single path per connection i.e TCP is more tightly tied to network path [1]. But modern devices increasingly have multiple network interfaces such as Wi-Fi, cellular, and Ethernet. Similarly modern data centers have redundant networks. Simultaneous use of these multiple paths for a TCP/IP session would improve resource usage within the network. Multipath TCP (MPTCP) has been introduced by the Internet Engineering Task Force (IETF) to use multiple paths effectively and efficiently between the sender and receiver. [1]. MPTCP is an extension of TCP (Transmission Control Protocol), which provides multiple paths, leading to higher throughput and low latency. Two major advantage of MPTCP is throughput aggregation and reliability [1].

Challenges. Although MPTCP has shown better performance than TCP in many applications, it has its own challenges. The network can become congested due to heavy traffic in the multiple paths (subflows) if the subflow rates are not determined correctly. Moreover, communication latency can occur if the packets are not scheduled correctly between the subflows. Hence scheduler is core of MPTCP , but heterogeneity of network makes efficient scheduling difficult. [2].

Existing approaches. Default scheduler in MPTCP known as minRTT, sends packets along the path with the lowest estimated RTT (round-trip time) until the congestion window of that path is full [2]. Then it shifts packets to the path with the next higher RTT. However, as reported in [3], minRTT suffers from low bandwidth utilization, particularly on heterogeneous paths. Other Existing schedulers such as BLEST, and ECF use network parameters and predefined heuristics to make scheduling decisions [3, 4]. BLEST, for example, attempts to reduce HoL blocking by estimating whether transmission over a slower path will cause receiver-side blocking [4]. Other adaptive data driven schedulers also exists based on reinforcement learning and other ml approaches [5].Recent research has also explored supervised machine learning for MPTCP scheduling and has produced datasets containing diverse scheduling parameters. However, the

ability of a supervised-learning-based scheduler to generalize from its training conditions to unseen and dynamically changing network conditions remains an important practical question.

Problem statement. To develop a lightweight supervised-learning-based MPTCP scheduler and investigate its ability to make effective scheduling decisions under heterogeneous network conditions. The study will specifically evaluate its generalization under unseen and dynamically changing network conditions.

Objectives. The objectives of the proposed work are:

(i) To build a supervised-learning-based MPTCP scheduler that decides how to split traffic across paths using network-state information such as RTT, bandwidth, loss rate, congestion window, and in-flight data.

(ii) To compare the learned scheduler with established MPTCP schedulers such as minRTT, BLEST, and ECF, and measure how well each one performs in terms of goodput, throughput, HoL blocking, retransmissions, and flow completion time.

(iii) and To investigate the generalization of the learned scheduler by testing it on network conditions that differ from those used during training, including dynamic changes in bandwidth, latency, and packet loss.

References

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